



SynBio Dictionary

miniGEM - 2025

Welcome to the SynBio dictionary: your guide to the fundamental vocabulary of synthetic biology.

Synthetic biology is new and multidisciplinary field. As such, you may not be completely familiar with its concepts, principles, or techniques. This dictionary seeks to introduce you to the key terms used by synthetic biologists to better equip you to explore the field. Every word comes with a phonetic spelling and definition, and some words come with a scale descriptor to indicate whether it acts on a molecular, cellular or organismal level.

The dictionary is separated into 5 different chapters.

Chapter One — the fundamental concepts in biology that synthetic biology most commonly works with

Chapter Two — the principles and techniques of engineering that synthetic biology incorporates in its practice

Chapter Three — the tools developed by synthetic biologists to solve real-world problems

Chapters Four and Five — the essential wet and dry lab techniques to develop and implement the tools



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Chapter One

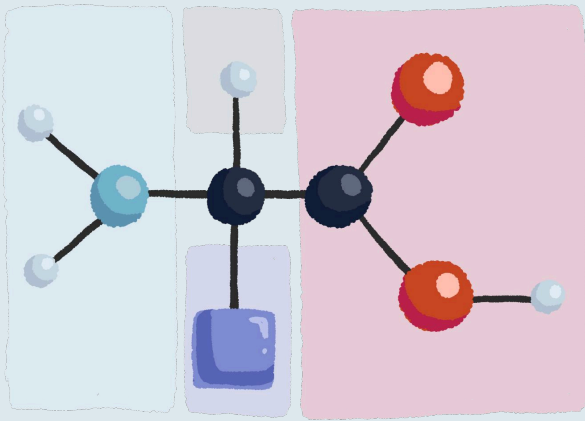
Fundamental Biology Concepts

Amino Acid

molecular

a·mi·no ac·id /ə'mēnō 'asəd/

An organic molecule that serves as the basic building block of proteins.



Anabolism

molecular

a·nab·o·lism /ə'nabə,lizəm/

A process in which living organisms use energy to produce complex molecules from simpler molecules.

Antibody

molecular

/'æn.ti,bɑ:di/

A protein produced by the immune system that recognizes and neutralizes foreign substances.

Antigen

molecular

/'æn.ti.dʒən/

A foreign substance in the body that causes an immune response such as the release of antibodies.

Biomolecule

mol.

/'baɪ.oʊ,mɑ:l.i.kju:l/

A molecule produced by living organisms essential to certain living processes. Examples include amino acids, carbohydrates, proteins, and lipids.

Carbohydrate

/ˌkɑːr.boʊˈhaɪ.dreɪt/

An organic molecule composed of carbon, hydrogen, and oxygen atoms. Primarily provides energy to organisms.

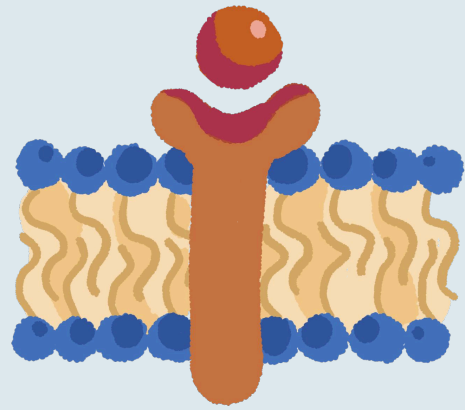
mol.

Cell surface receptor

/sɛl 'sɜːfɪs rɪ'sɛp.tə/

A protein on the surface of the cell that helps the cell communicate with its environment.

cellular



Catabolism

/kəˈtæb.əˌlɪz.əm/

A process in which living organisms release energy by breaking down complex molecules into simpler molecules.

molecular

Cell

/sɛl/

The smallest unit of life, made of a cytoplasm surrounded by a membrane.

cellular

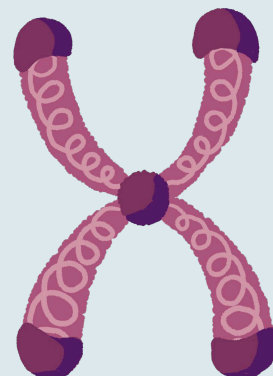


Chromosome

/'kroʊ.məˌsoʊm/

A structure containing the genetic material of an organism in the form of DNA wrapped around proteins (histones).

cellular



Codon

/ˈkoʊ.dɑːn/

molecular

A sequence of three nucleotides that encodes one amino acid.

Cytoplasm

/ˈsaɪ.təˌplæz.əm/

cellular

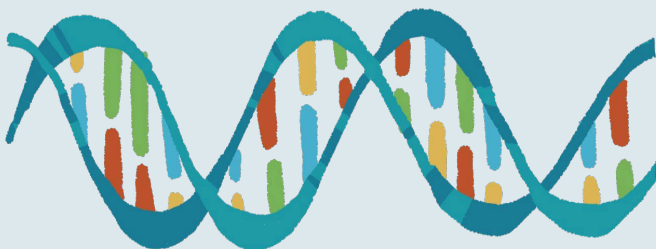
The fluid within a cell.

DNA

/ˌdiː.enˈeɪ/

molecular

Deoxyribonucleic acid: a double-stranded molecule made of a series of nucleotides that contains genetic information for making proteins.



DNA

replication

/ˌdiː.enˈeɪˌrɛp.liˈkeɪ.ʃən/

molecular

The process of making a copy of a DNA strand to give to a cell's daughter cell during mitosis



Emergent property

/ɪˈmɜː.dʒənt ˈprɒ.pə.ti/

A property that arises from the collective behaviours and interactions of the components of a system, but which does not belong to the individual components themselves.

Enzyme

/ˈɛn.zaɪm/

cellular

A protein that speeds up chemical reactions.

Eukaryote

/juːˈkær.i.oʊt/

organismal

Organisms composed of eukaryotic cells, such as plants, animals, fungi, and some microorganisms.

Eukaryotic cell

/juːˈkær.iˈɑː.tɪk sɛl/

cellular

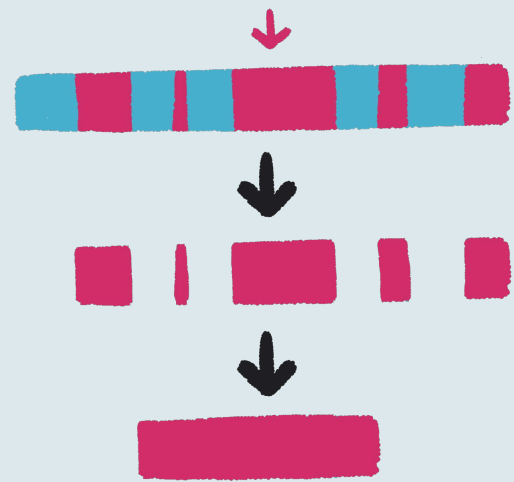
A cell that contains membrane-bound organelles and makes up eukaryotes.

Exon

/ˈɛk.sɑːn/

molecular

A nucleotide sequence in a gene that is not removed by RNA splicing before translation.



Extracellular matrix

/ɛkstrə ˈsɛljʊlər ˈmeɪtrɪks/

cell.

A structure of various molecules and proteins that exists between cells and gives structure to groups of cells.

Gene

/dʒiːn/

molecular

A series of nucleotides forming a sequence of DNA.

Genetics

molecular

/dʒəˈnɛ.tɪks/

The study of how genes are passed down between generations.

Genome

molecular

/'dʒiː.nəʊm/

The set of all of an organism's genes.

Hormone

molecular

/'hɔːr.məʊn/

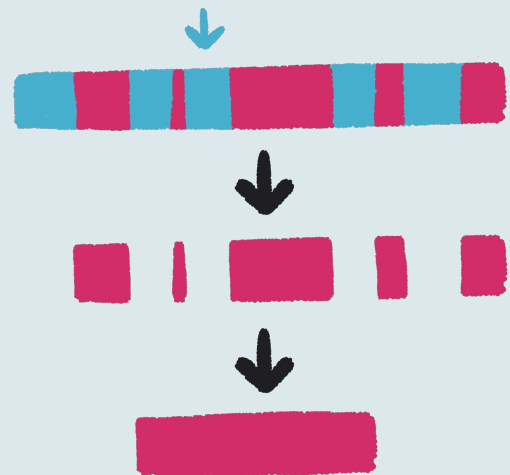
A molecule that sends signals throughout an organism's body.

Intron

molecular

/'ɪn.trən/

A nucleotide sequence in a gene that does not code for any protein and is removed before translation during RNA splicing.



Lipid

molecular

/'lɪ.pɪd/

A class of organic compounds including fats, oils, waxes, etc. They are often hydrophobic (repel water), but may be amphiphilic (partly repel water, partly attracted to water).

Mature RNA

/mə'tʃʊr ,ɑːr.en'eɪ/

RNA that has undergone post-transcriptional modification.

molecular

Membrane

/'mɛm.breɪn/

A thin layer of cell or tissue that acts as a barrier in an organism.

cellular

Membrane protein

/'mɛm.breɪn 'prəʊ.tiːn/

A protein that is embedded in or that interacts with the cell membrane.

molecular

Metabolic pathway

/,mɛt.ə'bɑː.lɪk 'pæθ.weɪ/

A connected series of chemical reactions that are either anabolic or catabolic occurring in an organism as part of its metabolism.

molecular

cellular

Metabolism

/mə'tæ.bə,lɪ.zəm/

The sum of all reactions occurring in an organism that convert its food into energy or into building blocks for its body.

molecular

cellular

Microorganism

/,maɪ.kroʊ'bɪr.gə,nɪ.zəm/

An organism that can only be seen through a microscope.

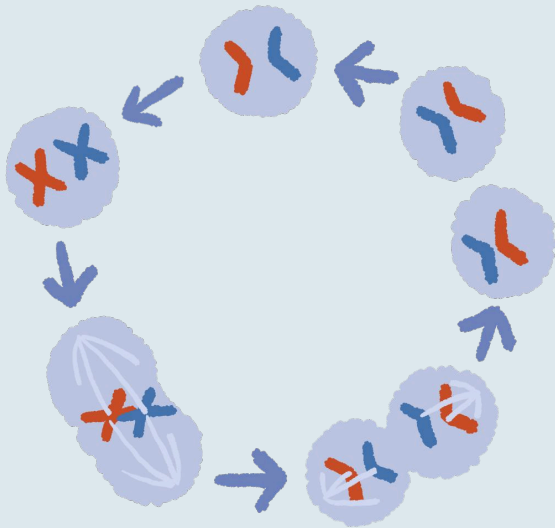
org.

Mitosis

/maɪ'toʊ.sɪs/

cellular

The process in which a cell divides into two genetically identical daughter cells.



mRNA

/ˌɛm.ɑːr.en'eɪ/

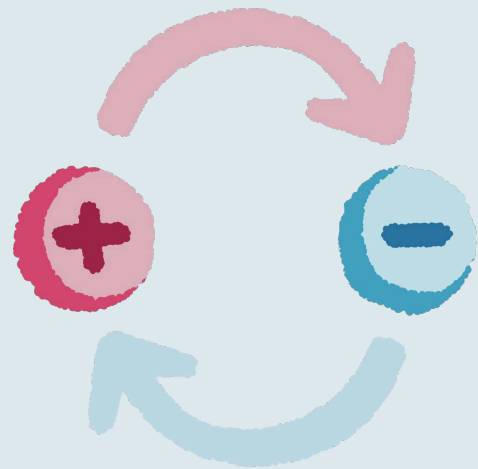
molecular

Messenger RNA: an RNA copy of DNA that is made during transcription and that will be read by a ribosome to be translated into a protein.

Negative feedback loop

/'nɛ.gə.tɪv 'fɪ.d.bæk luːp/

A cycle in which a change triggers a response that counteracts that change.



Nucleic acid

/nuː'kleɪ.ɪk 'æ.sɪd/

mol.

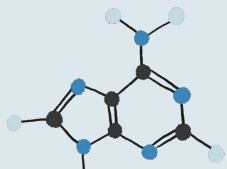
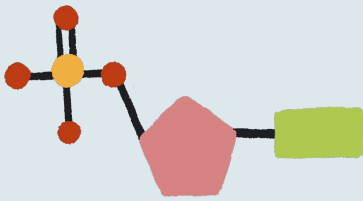
A large molecule, e.g. DNA or RNA, made of a series of nucleotides that carry genetic information for making proteins.

Nucleotide

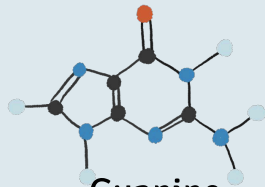
molecular

/ˈnuːkli.oʊ.taɪd/

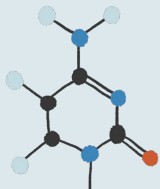
The basic building block of nucleic acids, made of a nitrogenous base, a pentose sugar, and a phosphate group.



Adenine



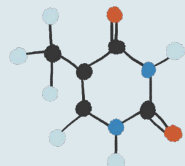
Guanine



Cytosine



Uracil



Thymine

● H ● C ● O ● N

Oligonucleotide

molecular

/ˌoʊ.li.goʊˈnuːkli.oʊ.taɪd/

A short single or double stranded piece of DNA or RNA.

Operon

molecular

/ˈɑː.pə.rɑːn/

A unit of DNA that controls gene expression.

Organelle

cellular

/ˌɔːr.gəˈneɪl/

A small structure within the cell that performs a specific function for it.

Peptide

molecular

/ˈpeɪ.taɪd/

A short sequence of amino acids (2-50).

Phosphorylation

molecular

/ˌfɑːs.fə.rəˈleɪ.ʃən/

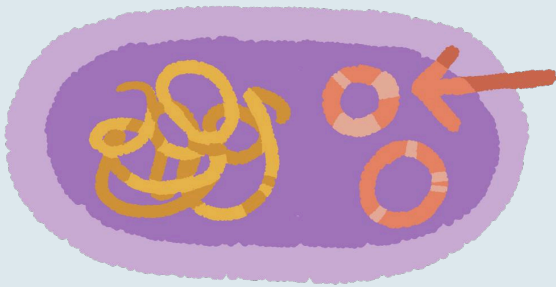
The addition of a phosphoryl (PO₃) group to a molecule or ion, which can activate or deactivate certain enzymes.

Plasmid

/ˈplæz.mɪd/

cellular

A small circular ring of DNA found in bacteria that provides them with additional genetic information and which replicates independently of them.



Polymerase

/pəˈlɪ.məˌreɪs/

molecular

A protein that makes long chains of molecules or nucleic acids.

Polypeptide chain

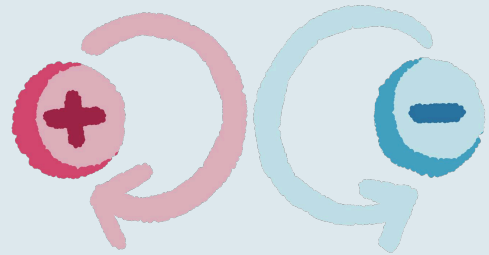
/ˌpɑː.liˈpɛp.taɪd tʃeɪn/

A sequence of amino acids, one or more of which make up a protein.

Positive feedback loop

/ˈpɑː.zə.tɪv ˈfiːd.bæk luːp/

A cycle in which a change triggers a response that furthers that change.



Post-transcriptional modification

/ˌpəʊst.trænˈskrɪp.ʃən.əl
ˌmɑː.də.fəˈkeɪ.ʃən/

mol.

A change to pre-mRNA after its transcription, such as RNA splicing.

Post-translational modification

/ˌpoʊst.trænz'leɪ.ʃən.əl
ˌmɑː.də.fə'keɪ.ʃən/

A change to a protein after it has been translated, such as phosphorylation.

Pre-mRNA

/priː ˌɛm.ɑːr.en'eɪ/

RNA produced during transcription that has yet to undergo post-transcriptional modification.

Prokaryote

/proʊ'kær.i.oʊt/

Organism composed of a single prokaryotic cell that has no nucleus or membrane-bound organelles.

mol.

Prokaryotic cell

/proʊ'kær.i'ɑː.tɪk sɛl/

A cell that has no nucleus or membrane-bound organelles.

cellular

Promoter

/prə'moʊ.tə/

A sequence of DNA to which protein binds to initiate the transcription of RNA.

molecular

Protein

/'proʊ.tiːn/

A molecule made of 1+ polypeptide(s) that performs a function in an organism.

molecular

Ribosome

/'raɪ.bə'soʊm/

An organelle that synthesizes proteins.

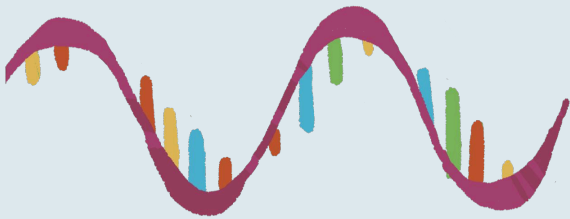
molecular

RNA

/ˌɑːr.en'eɪ/

molecular

Ribonucleic acid: a single-stranded molecule made of a series of nucleotides that has a variety of functions, including storing genetic information, and aiding in protein synthesis.

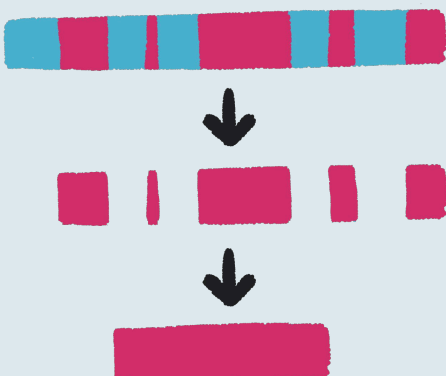


RNA splicing

/ˌɑːr.en'eɪ 'splaɪ.sɪŋ/

molecular

A process in which introns are removed from pre-RNA, and the remaining exons are joined together.



Signal transduction pathway

/'sɪɡ.nəl træns'dʌk.ʃən
'pæθ.weɪ/

molecular

cellular

A sequence of events in which molecules send signals to each other in a relay to perform some ultimate function.

Structural protein

/'strʌk.tʃə.əl 'prəʊ.tiːn/

molecular

A protein that provides shape and support to cells, tissues, and organisms such as collagen, keratin, and tubulin.

Substrate

/'sʌb.streɪt/

org. mol.

A substance on which an organism or an enzyme acts.

Terminator

molecular

/ˈtɜː.mə.neɪ.tə/

A sequence of nucleic acid that marks the end of a gene or operon during transcription.

Tissue

cellular

/ˈtɪʃ.uː/

A structure of cells and their extracellular matrix; groups of tissues make up organs.

Transcription

mol.

/trænˈskrɪp.ʃən/

The process by which an RNA copy called mRNA is made from a DNA sequence, to be used to make proteins.



Translation

molecular

/trænzˈleɪ.ʃən/

The process by which the information stored by the mRNA is used to combine amino acids in order to make proteins.



Transmembrane protein

mol.

/trænzˈmɛm.breɪn ˈprəʊ.tiːn/

aka integral protein; a protein that helps molecules move across the cell membrane.

Transport protein

molecular

/ˈtræns.pɔːrt ˈprɒʊ.tiːn/

A protein that helps molecules move across the cell membrane.

tRNA

molecular

/ˌtiː.ɑːr.enˈeɪ/

Transport RNA: a short sequence of RNA that aids in translation by connecting amino acids to their corresponding mRNA codon.

Chapter Two

Engineering Principles/Techniques

Abstraction

/æb'stræk.fən/

The principle of generalizing or hiding specific details to allow for focus on more important matters.

Bottom-up approach

/ˌbɒt.əm'ʌp ə'prɒʊtʃ/

An approach in which a larger system is created by putting together smaller systems.

Cell Engineering

/sɛl ,ɛn.dʒə'nɪr.ɪŋ/

The practice of modifying a cell's genes to change its functions.

cellular

Computer-aided design

/kəm'pjʊː.tə ,eɪ.dɪd dɪ'zaɪn/

A design of a product that has been developed, modified, analyzed or optimized by the use of computer software, either in 2D or 3D.

Genetic engineering

/dʒə'nɛ.tɪk ,ɛn.dʒə'nɪr.ɪŋ/

The practice of modifying genes to change an organism's function.

molecular

Metabolic engineering

/ˌmɛt.əˈbɔːlɪk ˌɛn.dʒəˈnɪr.ɪŋ/

Modifying an organism, often its genes, to induce it to produce or produce more of a desired substance.

molecular

Modularity

/ˌmɔː.dʒəˈlær.ə.ti/

The principle of designing components that can be separated and combined with other components in various ways to perform a variety of functions.

Protein engineering

/ˈprəʊ.tiːn ˌɛn.dʒəˈnɪr.ɪŋ/

Developing proteins that can perform desired functions, often by changing their amino acid sequence.

Standardization

/ˌstæn.də.dəˈzeɪ.ʃən/

The principle of following a pre-established guideline to ensure consistency.

Top-down approach

/ˌtɔːpˈdaʊn əˈprəʊtʃ/

An approach in which a larger system is understood by breaking it down into smaller components, which are each individually analyzed.

Chapter Three

Synthetic Biology

BioBrick

molecular

/ˈbaɪ.oʊ.brɪk/

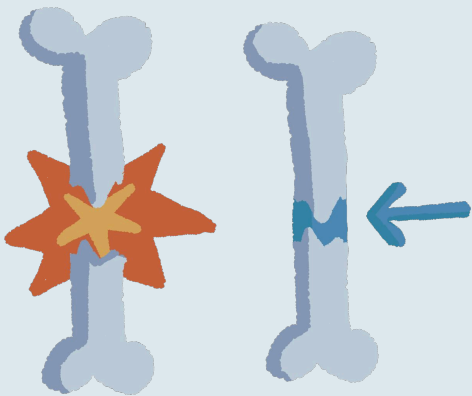
A sequence of DNA that has been isolated for synthetically building biological systems; they are modular and can be pieced together.

Biomaterial

cellular

/ˌbaɪ.oʊ.məˈtɪr.i.əl/

A substance that has been engineered to interact with the body for diagnostic or therapeutic medical purposes.



Biomimetics

cellular

/ˌbaɪ.oʊ.məˈmɛ.tɪks/

The study of imitating natural biological systems to solve human problems.

Biosensor

/ˈbaɪ.oʊ.sɛn.sə/

A device that detects chemical substances using a biological component such as an enzyme, cell or antibody.

Chassis

organismal

/ˈʃæ.si/

1. The framework of a manufactured structure.
2. The host organism that receives and uses a synthetic biological part.

DBTL cycle

/diː.biː.tiː'el 'saɪ.kəl/

Design-Build-Test-Learn cycle is an approach for developing solutions to synthetic biology problems.

Design



Build



Learn



Test

Fusion protein

/'fjuː.zən 'proʊ.tiːn/

Protein manufactured by joining together two genes that originally coded for different proteins.

Gene synthesis

mol.

/dʒiːn 'sɪn.θə.sɪs/

The process of manufacturing DNA strands without using a template strand by manually joining together nucleotides.

Genetically modified organism (GMO)

org.

/dʒə'nɛ.tɪ.kli 'mɑː.də.faɪd
'ɔːr.gənɪ.zəm/

An organism whose genome has been edited using genetic engineering.

Gene transplantation

molecular

/dʒiːn ,træns.plæn'teɪ.ʃən/

The process of replacing an organism's genome by inserting another organism's genome into it.

Minimal cell

/ˈmɪ.nə.məl sɛl/

A cell whose genome has been edited to only contain the genes needed for survival.

cell.

Regulatory circuits

/ˈrɛɡjəˌlɛtɔːi ˈsɜːkɪts/

Groups of genes that send signals to each other to influence each other's expressions.

molecular

Synthetic Biology Open Language (SBOL)

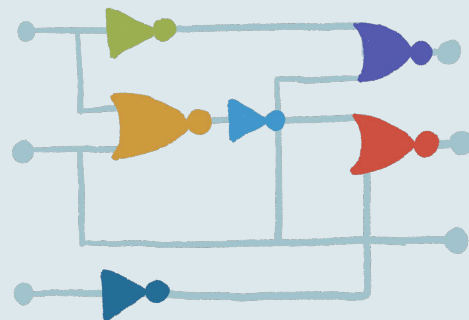
/sɪnˈθɛtɪk baɪˈɒlədʒi ˈoʊpən ˈlæŋɡwɪdʒ/

A standard for visually representing synthetic biology designs.

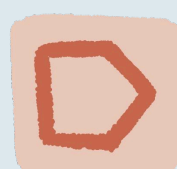
Synthetic circuit

/sɪnˈθɛtɪk ˈsɜːkɪt/

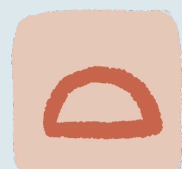
A group of biological parts that send signals and influence each other to perform logical functions, similar to an electric circuit.



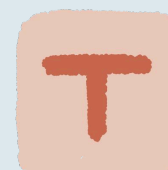
Promoter



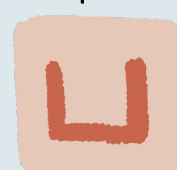
Coding sequence



Ribosome entry site



Terminator



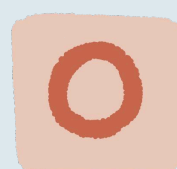
Operator



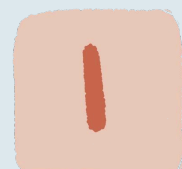
Insulator



Primer binding site



Origin of replication



Restriction site

Chapter Four

Wet Lab

Aseptic technique

/eɪ'sɛp.tɪk tɛk'niːk/

A procedure that maintains the sterility of lab equipment and prevents the spread of bacteria or contamination.

Assay

/'æ.seɪ/

A procedure that measures the presence, quantity, or activity of a substance, cell or organism.

Bacterial colony

/bæk'tɪr.i.əl 'kɑː.lə.ni/

A group of genetically identical bacteria, often descending from a single parent bacterium.

cellular

Cas9

/kæs 'naɪn/

molecular

An enzyme that is able to open up DNA sequences complementary to CRISPR sequences.

Cell culture

/sɛl 'kʌl.tʃə/

cellular

The process of growing cells in the lab.

Cell-free system

/sɛl 'friː 'sɪs.təm/

A system in which biological reactions that normally occur in cells are tested outside of the cell, in a synthetic environment.

Chromatography

/ˌkroʊ.məˈtɑː.grə.fi/

A lab technique in which a mixture is separated into its components (e.g. column chromatography, paper chromatography, etc.)

Cloning

/'kloʊ.nɪŋ/

The process of producing genetically identical organisms.

organismal

CRISPR

/'krɪs.pə/

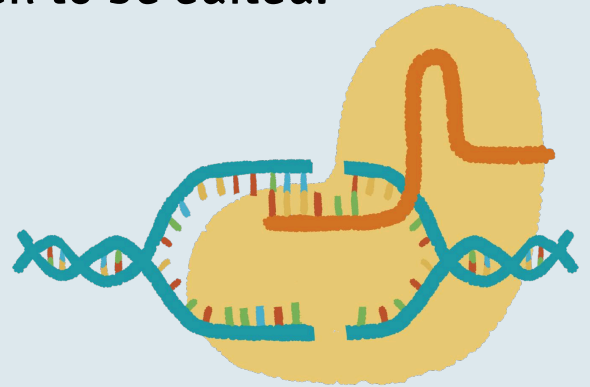
Clustered regularly interspaced short palindromic repeats: a family of DNA sequences found in some prokaryotes genomes.

molecular

CRISPR-Cas9

/'krɪs.pə kæs 'naɪn/

A genetic engineering technique in which the genes of an organism can be cut open to be edited.



mol.

DNA sequencing

/ˌdiː.en'eɪ 'siː.kwən.sɪŋ/

The process of determining the nucleotide sequence of DNA

mol.

E. coli

/iː 'koʊ.laɪ/

A model organism for molecular biology research due to it being easy to culture.

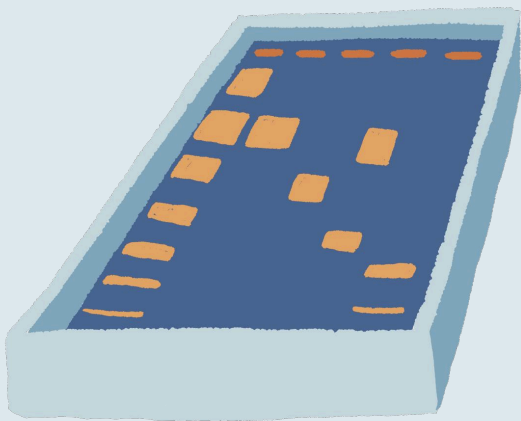
cellular

Gel electrophoresis

molecular

/dʒel ɪˌlek.trə.fəˈriː.sɪs/

A lab technique used to sort proteins, DNA or RNA by molecular size.



Gene editing

molecular

/dʒiːn ˈɛ.dɪ.tɪŋ/

A group of lab techniques that modify an organism's genes.

In vitro

/In ˈviː.troʊ/

Experiments that are performed outside of an organism, in lab equipment such as test tubes.

In vivo

/In ˈviː.vəʊ/

Experiments that are performed in an organism.

Model organism

cellular

/'mɔː.dəl ˈɔːr.gə.nɪ.zəm/

A non-human species that is used to study specific biological phenomena, often with the belief that this knowledge will apply to other species. Model organisms may be picked for their biological similarity to humans, or due to them being easy to work with.

Petri dish

/ˈpiːtri dɪʃ/

A shallow, lidded plastic dish often filled with a growth-promoting medium on which cells can be cultured.

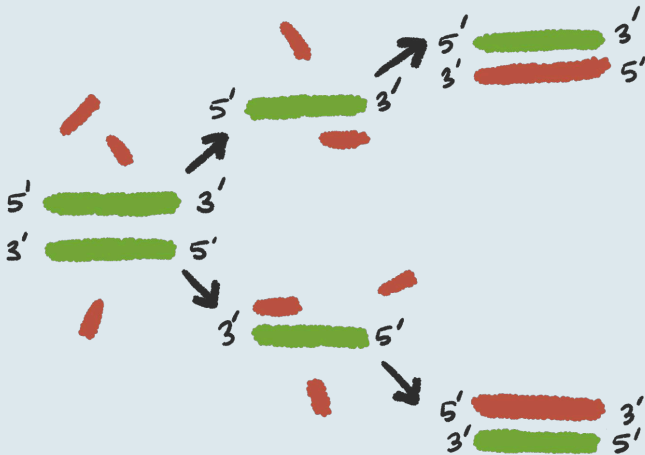


Polymerase chain reaction

molecular

/pəˈliːməˌreɪs tʃeɪn riˈæk.ʃən/

A lab technique performed to replicate DNA into a large enough quantity to be studied.



S. cerevisiae

cell.

/ˌsæk.kəˈroʊ.my.sɪz ˌser.əˈviː.i.ə/

A model organism for studying eukaryotes.

Transformation

mol.

cell.

/ˌtræns.fəˈmeɪ.ʃən/

A process in which a cell's genes are altered by uptaking genetic material from its environment.

Western blotting

molecular

/ˈwɛs.tən ˈblɑː.tɪŋ/

A lab technique that detects the presence of a protein in a sample by separating them by size and identifying them with antibodies.

Chapter Five

Dry Lab

Agent-based modeling

/ˈeɪ.dʒənt beɪst ˈmɑː.dəl.ɪŋ/

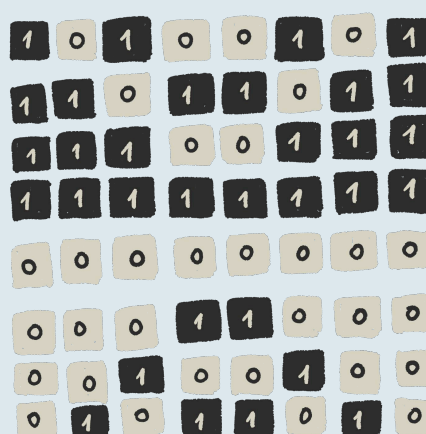
A computer simulation where “agents” are encoded with pre-defined rules on how they behave and interact with other agents and their environment. For example, an agent-based model might study how cells interact with other cells during injury. A type of cellular automaton model.

Cellular automaton

cellular

/ˈsɛl.jə.lə ɔːˈtɑː.mə.tən/

A model of computation consisting of a grid of cells that each have a state that is affected by both time and the state of neighbouring cells.



Computational fluid dynamics model

/ˌkɑːm.pjə'teɪ.ʃən.əl 'fluːɪd daɪ'næ.mɪks 'mɑː.dəl/

A model that predicts and studies the flow of liquids and gases.

Deterministic model

/dɪ'tɜː.mə'nɪs.tɪk 'mɑː.dəl/

A model whose output is entirely dependent on its input, and does not involve randomness.

Differential model

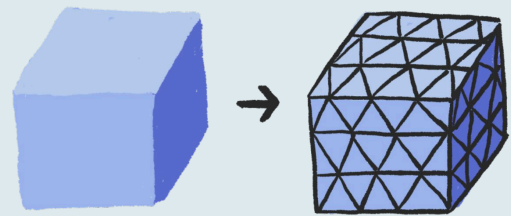
/ˌdɪ.fə'ren.ʃəl 'mɑː.dəl/

A model that studies how variables change over time.

Finite element method

/'faɪ.naɪt 'ɛ.lə.mənt 'mɛ.θəd/

A method for solving physics and engineering problems involving complex systems by dividing the system into a finite number of simpler parts.



In silico

/ɪn 'sɪ.lɪ.koʊ/

Experiments that are performed on a computer.

Machine learning model

/mə'ʃiːn 'lɜː.nɪŋ 'mɑː.dəl/

A computer program that can recognize patterns to make predictions or decisions about unforeseen data.

Mathematical model

/ˌmæθ.əˈmæt.ɪ.kəl ˈmɑː.dəl/

A model that represents real-world situations or problems using mathematical concepts.

Molecular dynamics

/məˈlɛk.jə.lə daɪˈnæ.mɪks/

A computer simulation that predicts the movement of atoms and molecules.

Multiscale model

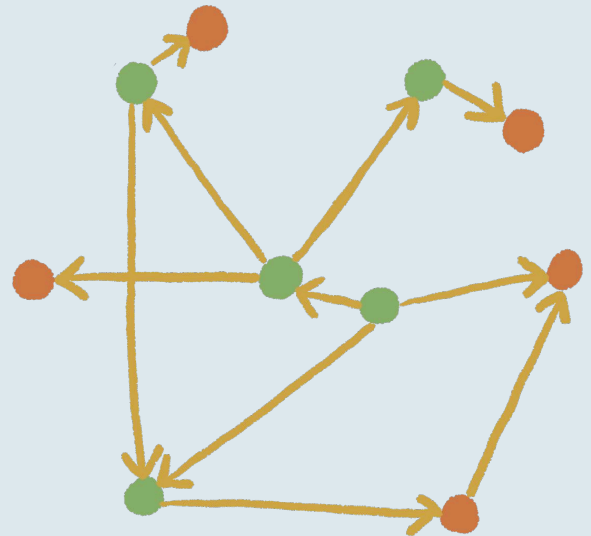
/ˈmʌl.ti.skeɪl ˈmɑː.dəl/

A model that represents multiple scales, such as modeling the physics of fluid flow and the behaviour of cells simultaneously.

Network model

/ˈnɛt.wɜːk ˈmɑː.dəl/

A model that represents real-world situations or problems using graphs consisting of nodes and vertices.



Stochastic model

/stɒˈkæ.stɪk ˈmɑː.dəl/

A model whose output is not entirely determined by its input, but factors in randomness.